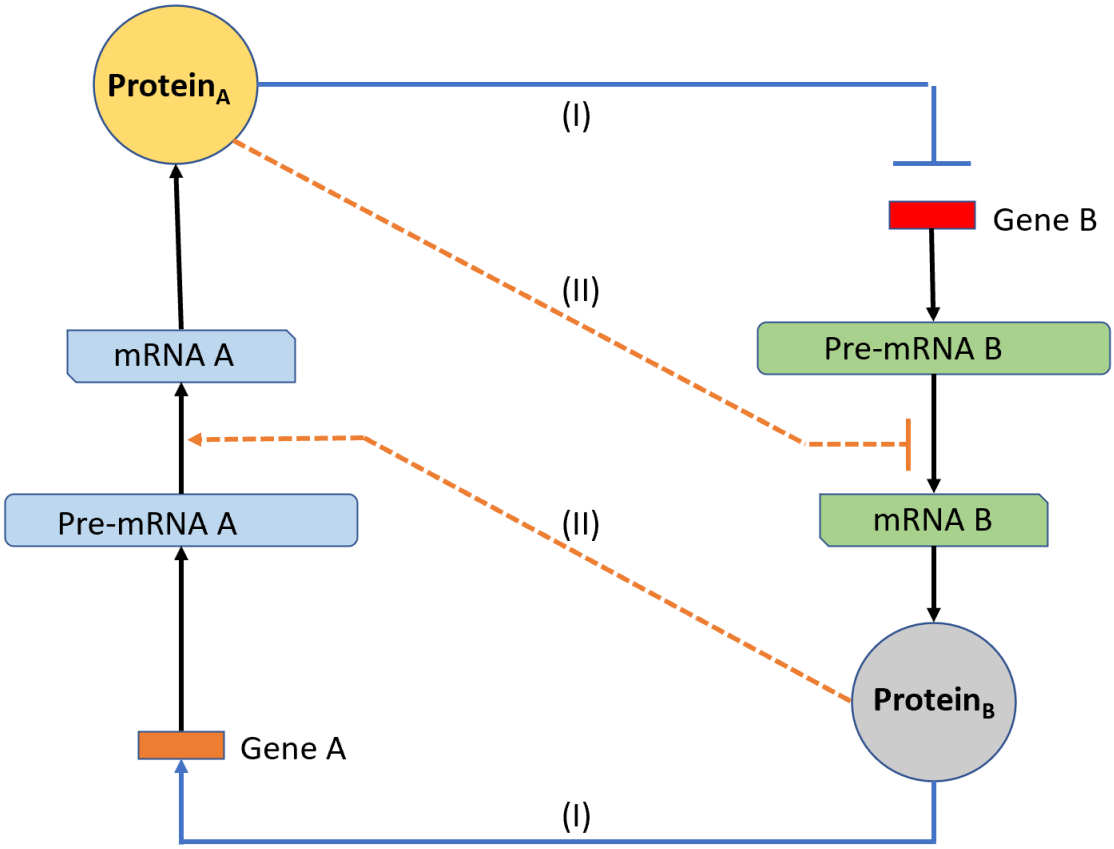

Solid blue line -Transcription Factor – influences transcription

Orange dotted lines - Splicing Factor – influences splicing

Figure 1: Gene interaction network

Project Chronos: Deciphering the Dynamics of Cancer Gene Regulation

Mission Briefing

Welcome to the Chronos Initiative, a cutting-edge biodynamics research group. Your team has been tasked with a critical mission: to understand the molecular machinery behind a particularly aggressive form of liver cancer.

Our genomics division has identified a key gene regulatory network involving two genes:

Gene A (*GUARDIAN*): This gene codes for a tumor-suppressor protein, **Protein A**. Its job is to keep cell growth in check.

Gene B (*PROLIFERATOR*): This gene codes for an oncogenic protein, **Protein B**. When overexpressed, it drives uncontrolled cell division.

These two proteins regulate each other in a delicate dance as shown in interaction map (figure 1).

Our preliminary data suggests that in cancer cells, Protein A's ability to suppress Protein B is discrupted. This disruption can occur via one of two distinct mechanisms:

Mechanism I (**Transcriptional Hijack**): Protein A normally acts as a transcription factor, binding to the PROLIFERATOR gene's promotor to block its transcription. In the less aggressive, early-stage cancer, this transcriptional repression is disabled. The cell has found a way to prevent Protein A from binding to the Gene B promoter, but Protein A's other functions may remain intact.

Mechanism II (**Splicing Sabotage**): Protein A normally acts as a splicing factor. interfering with the maturation of PROLIFERATOR pre-mRNA into functional mRNA. In the highly aggressive, fast-growing cancer, this splicing inhibition by Protein A is the dominant active mechanism causing the disruption, leading to a massive buildup of unprocessed pre-mRNA.

Your mission is to analyze total RNA-sequencing data from two different patient-derived cell lines to determine which mechanism is active in each and to model the resulting cellular dynamics.

Task 1: Analysis of Patient Sample Alpha

Your team has just received data from the first cell line, "Patient Alpha." This sample is believed to represent a less aggressive, early-stage, form of the cancer. The total RNA-seq data for the PROLIFERATOR gene shows a high abundance of reads matching the sequence **AGCGC**.

(a) Identify the Regulatory Mechanism: Analyze the sequence AGCGC to infer which regulatory mechanism (I or II) is active in this cell line.

* Hint: The classification of the total RNA-sequence data should be carried out using the Viterbi algorithm. Use the HMM parameters specified in Tables 1, 2, and 3.

(b) Based on the mechanism you identified for patient alpha, choose the appropriate mathematical framework to model the gene regulation dynamics ODE model vs SDEVelo model. Also, the choice of model formulation may depend on the observed mRNA status. Parameter values for these models are given in Tables 4 and 5, respectively. Ensure to present relevant graphs/visualizations.

Task 2: Analysis of Patient Sample Beta

Next analyze "Patient Beta", a cell line derived from a highly aggressive and fast-growing tumor. The RNA-seq data for the PROLIFERATOR gene in this sample is dominated by the sequence **AUUAU**. Repeat the full analysis pipeline (a) and (b) from Patient Alpha for this new sample.

Task 3: Comparative Analysis & Diagnosis

Synthesize your findings from Patient Alpha and Patient Beta to deliver a diagnosis.

Compare the protein phase portraits for Patient Alpha and Patient Beta. Describe the dynamic behaviors observed in each plot. What do these different behaviors imply about the cellular "fate" of each sample? Justify your conclusions based on the trajectories in your plots.

Table 1:

Transition probabilities	**Exon**	**Intron**
Exon	0.9	0.1
Intron	0.2	0.8

Table 2:

Emission Probabilities	**A**	**U**	**G**	**C**
Exon	0.25	0.25	0.25	0.25
Intron	0.4	0.4	0.05	0.15

Table 3:

Initial Probabilities
∏(Exon)=0.5
∏(Intron) = 0.5

Table 4:

SDEVelo parameters
a_A = 1.0 s
a_B = 0.25 s
b_A = 0.0005
b_B = 0.0005
c_A = 2.0 Ms^{-1}
c_B = 0.5 Ms^{-1}
β_A = 2.35 s^{-1}
β_B = 2.35 s^{-1}
γ_A = 1.0 s^{-1}
γ_B = 1.0 s^{-1}
$n_A = n_B = 3$
θ_A = 0.21 M
θ_B = 0.21 M
k_{PA} = 1.0 s^{-1}
k_{PB} = 1.0 s^{-1}
m_A = 2.35 s^{-1}
m_B = 2.35 s^{-1}
δ_{PA} = 1.0 s^{-1}
δ_{PA} = 1.0 s^{-1}
σ_{1A} = 0.05 $Ms^{-1/2}$
σ_{2A} = 0.05 $Ms^{-1/2}$

σ_{1B} = 0.05 $Ms^{-1/2}$
σ_{2B} = 0.05 $Ms^{-1/2}$
Initial concentration of unspliced mRNA A (pre mRNA)= 0.8 M
Initial concentration of unspliced mRNA B (pre mRNA) = 0.8 M
Initial concentration of protein A = 0.8 M
Initial concentration of protein B = 0.8 M
Initial concentration of spliced mRNA A = 0.8 M
Initial concentration of spliced mRNA B = 0.8 M

Table 5:

ODE parameters
m_A=2.35 s^{-1} # max Transcription rate of Gene A
m_B=2.35 s^{-1} # max Transcription rate of Gene B
γ_A = 1 s^{-1} # mRNA A deg rate
γ_B = 1 s^{-1} # mRNA B deg rate
k_{PA} = 1.0 s^{-1} # Translation rate of Protein A
k_{PB} = 1.0 s^{-1} # Translation rate of Protein B
θ_A = 0.21 M # expression threshold Protein_A binding
θ_B = 0.21 M # expression threshold for Protein_B binding
n_A = 3 # Hill coefficient for $Protein_A$ (increased for stronger nonlinearity)
n_B = 3 # Hill coefficient for $Protein_B$ (increased for stronger nonlinearity)
δ_{PA} = 1.0 s^{-1} # Degradation rate of $Protein_A$

δ_{PB} = 1.0 s^{-1} # Degradation rate of $Protein_B$
Initial concentration of mRNA A = 0.8 M
Initial concentration of mRNA B = 0.8 M
Initial concentration of protein A = 0.8 M
Initial concentration of protein B = 0.8 M

2. (Bonus) **Modeling Downstream Metabolic Effects**

A collaborating metabolic research team is interested in the downstream consequences. They hypothesize that Protein B (PROLIFERATOR), doesn't act alone it activates a key metabolic enzyme (E) that consumes a cellular resource (R) to fuel rapid growth of cancer. They've proposed a simplified model for this interaction:

$$\frac{dR}{dt} = \alpha R - \beta RE$$

$$\frac{dE}{dt} = -\gamma E + \delta RE$$

Here R is concentration of a metabolite, E is concentration of the growth promoting enzyme.

Your final mission tasks are:

a) Explain what each term in the equations (αR, −βRE, −γE, δRE) represents biologically in the context of a cellular resource and a growth-promoting enzyme. (In your Model/Eqn video)
b) Carry out a stability analysis for, $\alpha = 2$, $\beta = 1.1$, $\gamma = 1$ and $\delta = 0.9$, R(0) = 1 and E(0) =0.5 (And discuss in your results and discussions video)
c) Plot the stream plot with equilibrium point and nullclines (And explain in your Figures video)